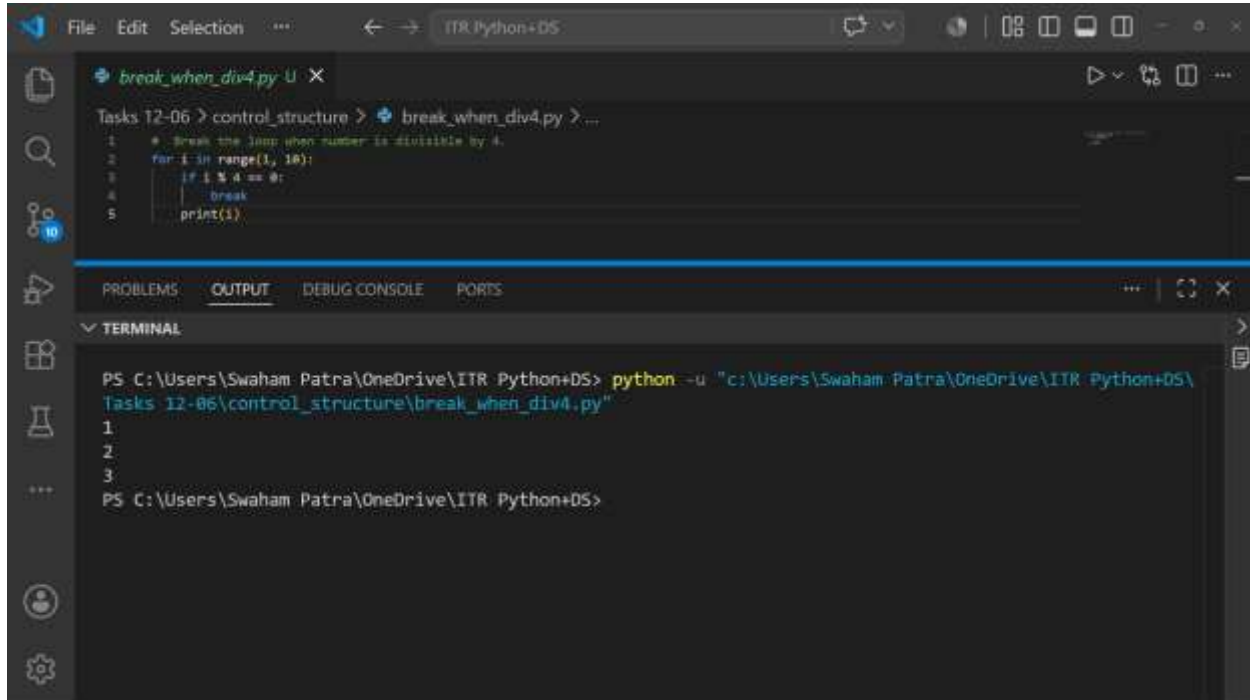


Swaham Pitambar Patra

Control Structure

Break the loop when number is divisible by 4.



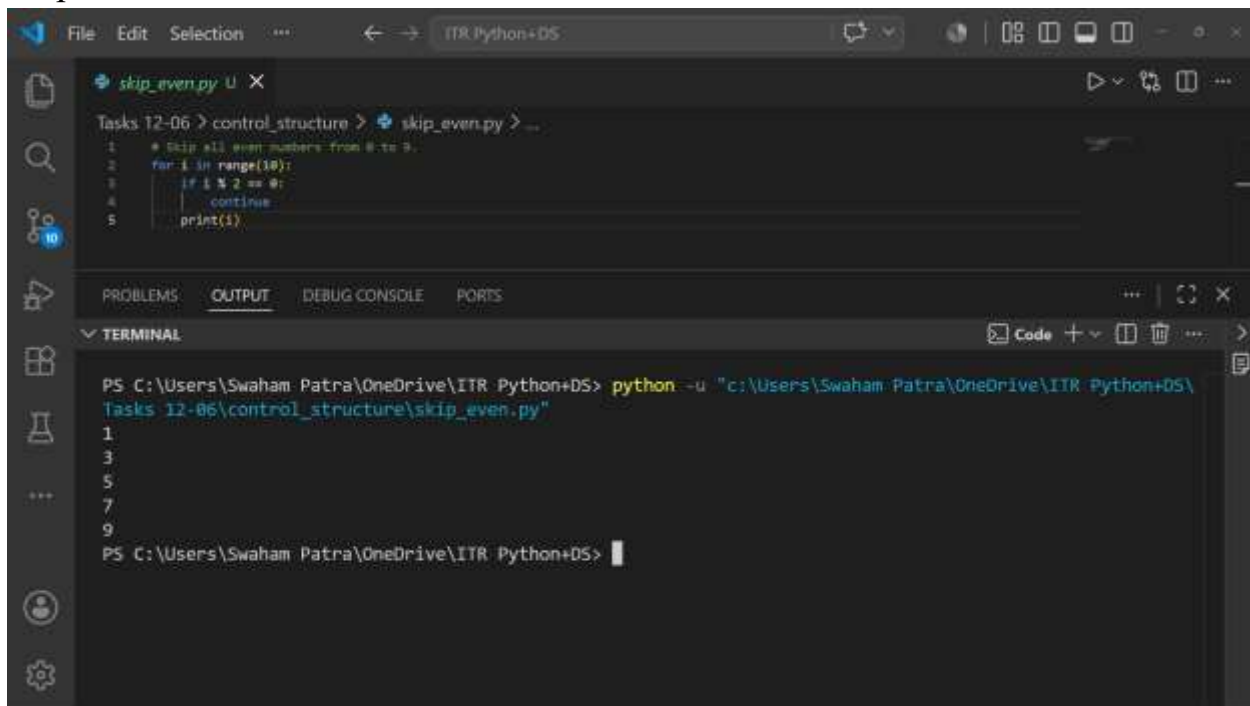
The screenshot shows the Visual Studio Code editor with a file named `break_when_div4.py` open. The code is as follows:

```
1 # Break the loop when number is divisible by 4.
2 for i in range(1, 10):
3     if i % 4 == 0:
4         break
5     print(i)
```

The `OUTPUT` pane shows the execution results:

```
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS> python -u "c:\Users\Swaham Patra\OneDrive\ITR Python+DS\
Tasks 12-06\control_structure\break_when_div4.py"
1
2
3
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS>
```

Skip all even numbers from 0 to 9.



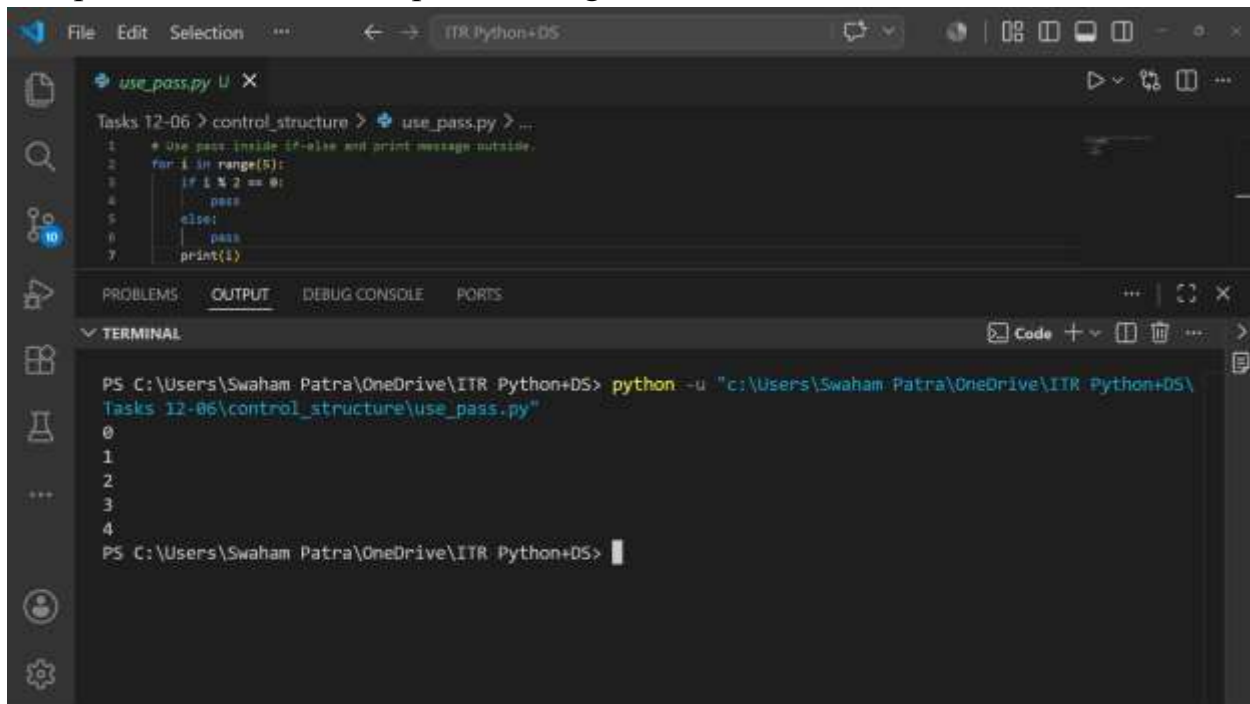
The screenshot shows the Visual Studio Code editor with a file named `skip_even.py` open. The code is as follows:

```
1 # Skip all even numbers from 0 to 9.
2 for i in range(10):
3     if i % 2 == 0:
4         continue
5     print(i)
```

The `OUTPUT` pane shows the execution results:

```
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS> python -u "c:\Users\Swaham Patra\OneDrive\ITR Python+DS\
Tasks 12-06\control_structure\skip_even.py"
1
3
5
7
9
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS>
```

Use pass inside if-else and print message outside.



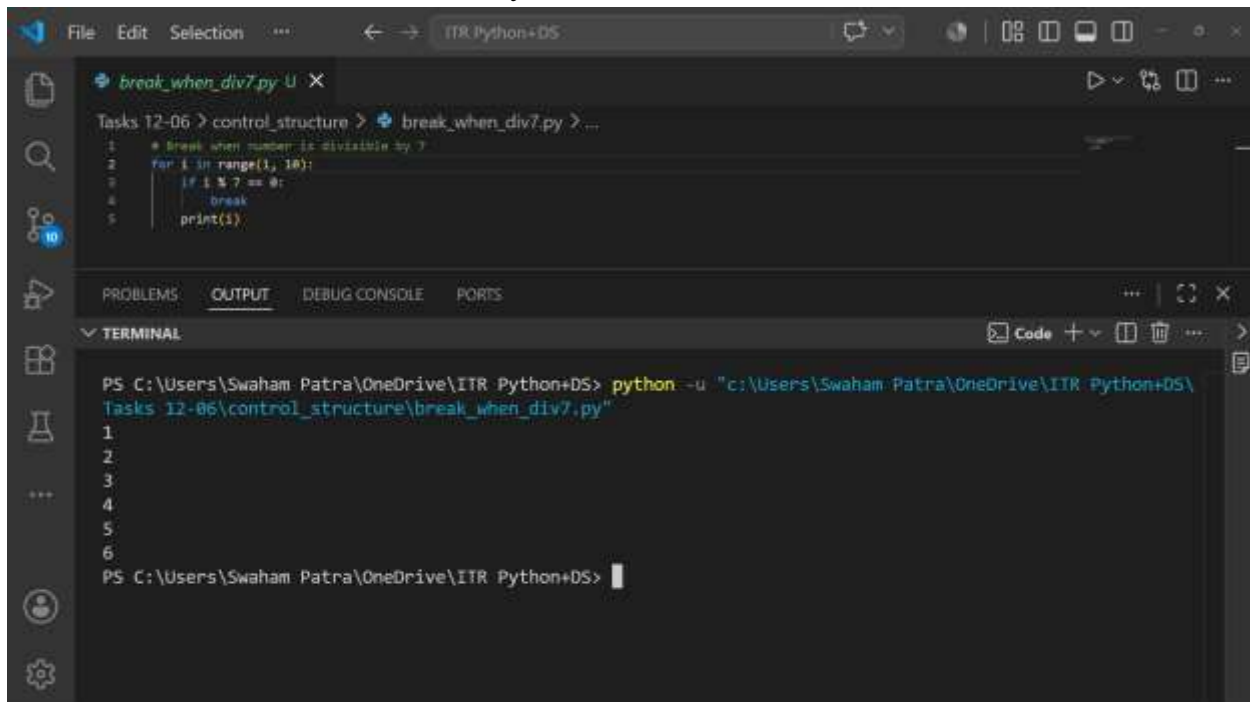
The screenshot shows the Visual Studio Code editor with a file named `use_pass.py` open. The code in the file is as follows:

```
1 # Use pass inside if-else and print message outside.
2 for i in range(5):
3     if i % 2 == 0:
4         pass
5     else:
6         pass
7 print(i)
```

The terminal window at the bottom shows the command `python -u "c:\Users\Swaham Patra\OneDrive\ITR Python+DS\Tasks 12-06\control_structure\use_pass.py"` being executed. The output of the program is:

```
0
1
2
3
4
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS>
```

Break when number is divisible by 7



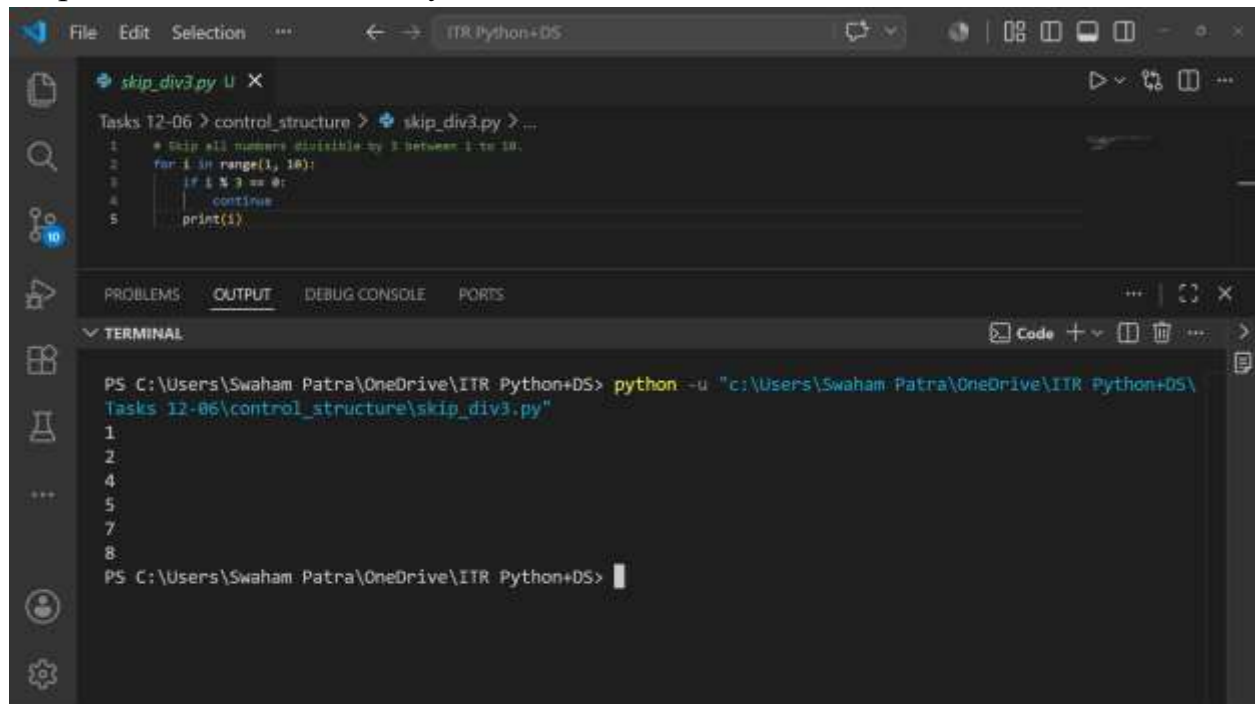
The screenshot shows the Visual Studio Code editor with a file named `break_when_div7.py` open. The code in the file is as follows:

```
1 # Break when number is divisible by 7
2 for i in range(1, 10):
3     if i % 7 == 0:
4         break
5     print(i)
```

The terminal window at the bottom shows the command `python -u "c:\Users\Swaham Patra\OneDrive\ITR Python+DS\Tasks 12-06\control_structure\break_when_div7.py"` being executed. The output of the program is:

```
1
2
3
4
5
6
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS>
```

Skip all numbers divisible by 3 between 1 to 10.



The image shows a Visual Studio Code editor window with a Python file named `skip_div3.py` open. The code in the editor is as follows:

```
1 # Skip all numbers divisible by 3 between 1 to 10.  
2 for i in range(1, 10):  
3     if i % 3 == 0:  
4         continue  
5     print(i)
```

Below the editor, the **TERMINAL** panel is active, showing the command `python -u "c:\Users\Swaham Patra\OneDrive\ITR Python+DS\Tasks 12-06\control_structure\skip_div3.py"` and its output:

```
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS> python -u "c:\Users\Swaham Patra\OneDrive\ITR Python+DS\Tasks 12-06\control_structure\skip_div3.py"  
1  
2  
4  
5  
7  
8  
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS>
```

The output shows the numbers 1, 2, 4, 5, 7, and 8, which are the numbers from 1 to 10 that are not divisible by 3.

Data structure

List

Code:

```
File Edit Selection View Go Run - Python 3.8
Tasks 12.06 > data structure > list.py -
1 # Create a list of your favorite colors and print it.
2
3 # Create a list of your favorite animals and print it.
4 list_of_animals = ['lion', 'tiger', 'elephant']
5 print("My favorite animals: ", list_of_animals)
6
7 # Create a list of three numbers, print the third number using indexing.
8 numbers = [10, 20, 30]
9 print("The original list of numbers: ", numbers)
10 print("The third number is: ", numbers[2])
11
12 # Create a list of five animals and print the animals from using slicing.
13 animals = ['lion', 'tiger', 'elephant', 'giraffe', 'zebra']
14 print("The original list of animals: ", animals)
15 print("The middle three animals are: ", animals[1:4])
16
17 # Create a list of four items, print the second to last item using negative indexing.
18 items = ['apple', 'banana', 'cherry', 'date']
19 print("The original list of items: ", items)
20 print("The second to last item is: ", items[-2])
21
22 # Change the last element of a list numbers = [10, 20, 30] to 40 and print the list.
23 numbers = [10, 20, 30]
24 print("The original list is: ", numbers)
25 numbers[-1] = 40
26 print("The updated list is: ", numbers)
27
28 # Create with numbers = [1, 2, 3], the append to add 4 and print the list.
29 numbers = [1, 2, 3]
30 print("The original list is: ", numbers)
31 numbers.append(4)
32 print("The updated list is: ", numbers)
33
34 # Create letters = ['a', 'c', 'd'], insert 'b' at index 1 and print the list.
35 letters = ['a', 'c', 'd']
36 print("The original list is: ", letters)
37 letters.insert(1, 'b')
38 print("The updated list is: ", letters)
39
40 # Create nums = [1, 2, 3, 4], remove the number 3 and print the list.
41 nums = [1, 2, 3, 4]
42 print("The original list is of nums: ", nums)
43 del nums[2]
44 print("The updated list is of nums: ", nums)
```

Output:

```
File Edit Selection View Go Run - Python 3.8
PROBLEMS OUTPUT DEBUG CONSOLE PORTS
TERMINAL
The third number is: 30
The original list of animals: ['lion', 'tiger', 'elephant', 'giraffe', 'zebra']
The middle three animals are: ['tiger', 'elephant', 'giraffe']
The original list of items: ['apple', 'banana', 'cherry', 'date']
The second to last item is: cherry
The original list is: [10, 20, 30]
The updated list is: [10, 20, 40]
The original list is: [1, 2, 3]
The updated list is: [1, 2, 3, 4]
The original list is: ['a', 'c', 'd']
The updated list is: ['a', 'b', 'c', 'd']
The original list is of nums: [1, 2, 3, 4]
The updated list is of nums: [1, 2, 4]
PS C:\Users\Swaham Patra\OneDrive\ITR Python\DS> []
```

Tuples

Code:

```
1 # Create a tuple with three elements: (1, 2, 3) and print it.
2
3 tuple_of_numbers = (1, 2, 3)
4 print("My tuple of numbers: ", tuple_of_numbers)
5
6 # Given a tuple (1, 2, 3, 4, 5), print the third element.
7 t = (1, 2, 3, 4, 5)
8 print("The original tuple is: ", t)
9 print("The third element is: ", t[2])
10
11 # Given a tuple of four colors, print the second to last color using negative indexing.
12 colors = ('red', 'blue', 'green', 'yellow')
13 print("The original tuple of colors is: ", colors)
14 print("The second to last color is: ", colors[-2])
15
16 # Slice the tuple (100, 200, 300, 400, 500) to get (200, 300)
17 sliced_tuple = (100, 200, 300, 400, 500)[1:3]
18 print("The original tuple is: ", (100, 200, 300, 400, 500))
19 print("The sliced tuple is: ", sliced_tuple)
20
21 # Create a tuple (2, 3, 4, 5, 6, 7, 8) and print the number of times 2 appears.
22 tuple_with_duplicates = (2, 3, 2, 2, 2, 4)
23 print("The original tuple is: ", tuple_with_duplicates)
24 print("The number of times 2 appears is: ", tuple_with_duplicates.count(2))
25
26 # Join (7, 8) and (9, 10) into one tuple and print it.
27 tuples = (7, 8)
28 print("The first tuple is: ", tuples)
29 tuple2 = (9, 10)
30 print("The second tuple is: ", tuple2)
31 joined_tuple = tuples + tuple2
32 print("The joined tuple is: ", joined_tuple)
```

Output:

```
PS C:\Users\Saaham Patra\OneDrive\ITR Python\DS> python -u "c:\Users\Saaham Patra\OneDrive\ITR Python\DS\Tasks 12-06\data_structures\tuple.py"
My tuple of numbers: (1, 2, 3)
The original tuple is: (1, 2, 3, 4, 5)
The third element is: 4
The original tuple of colors is: ('red', 'blue', 'green', 'yellow')
The second to last color is: green
The original tuple is: (100, 200, 300, 400, 500)
The sliced tuple is: (200, 300)
The original tuple is: (2, 3, 2, 2, 2, 4)
The number of times 2 appears is: 5
The first tuple is: (7, 8)
The second tuple is: (9, 10)
The joined tuple is: (7, 8, 9, 10)
PS C:\Users\Saaham Patra\OneDrive\ITR Python\DS>
```

Dictionary

Code:

```
File Edit Selection View Go Run Python 3.8
dictionary.py
Tasks 12-06 > data_structure > dictionary > _
1 # Dictionary
2
3 # Create a dictionary for a book with keys title, author, and year, then print it.
4 book = {'title': 'Dead Person's Viewpoint', 'author': 'Swahen', 'year': 2006}
5 print("The book dictionary is: ", book)
6
7 # Create a dictionary for a student with keys name, grade, and age, then print it.
8 student = {'name': 'A', 'grade': 'A', 'age': 18}
9 print("The student's grade is: ", student['grade'])
10
11 # Create a dictionary for a student with keys name, grade, and age, then print it.
12 student = {'name': 'A', 'grade': 'A', 'age': 18}
13 print("The updated dictionary is: ", student)
14
15 # Create a dictionary for a student with keys name, grade, and age, then print it.
16 student = {'name': 'A', 'grade': 'A', 'age': 18}
17 student['age'] = 19
18 print("The updated dictionary is: ", student)
19
20 # Create a dictionary for a student with keys name, grade, and age, then print it.
21 student = {'name': 'A', 'grade': 'A', 'age': 18}
22 student['age'] = 19
23 print("The updated dictionary is: ", student)
24
25 # Create a dictionary for a student with keys name, grade, and age, then print it.
26 student = {'name': 'A', 'grade': 'A', 'age': 18}
27 student['age'] = 19
28 print("The keys are: ", student.keys())
29 print("The values are: ", student.values())
```

Output:

```
File Edit Selection View Go Run Python 3.8
PROBLEMS OUTPUT DEBUG CONSOLE PORTS
TERMINAL
PS C:\Users\Swahen Patra\OneDrive\ITR Python\DS> python -u "c:\Users\Swahen Patra\OneDrive\ITR Python\DS\Tasks 12-06\data_structure\dictionary.py"
The book dictionary is: {'title': 'Dead Person's Viewpoint', 'author': 'Swahen', 'year': 2006}
The student's grade is: A
The updated dictionary is: {'name': 'A', 'grade': 'A', 'age': 18}
The updated dictionary is: {'name': 'A', 'grade': 'A', 'age': 19}
The updated dictionary is: {'name': 'A', 'grade': 'A', 'age': 19}
The keys are: ['name', 'grade', 'age']
The values are: ['A', 'A', 19]
PS C:\Users\Swahen Patra\OneDrive\ITR Python\DS>
```

Sets

Code:

```
File Edit Selection View Go Run Python 3.8
Tasks 12-06 > data_structure > set.py
1 # Set
2
3 # Create a set of three letters, e.g., {'a', 'b', 'c'} and print it.
4 a = {'a', 'b', 'c'}
5 print("The set is: ", a)
6
7 # Now, add a = {'x', 'y'} and 'c' to the set and print it.
8 a = {'a', 'y'}
9 a.add('c')
10 print("The updated set is: ", a)
11
12 # Given a = {'x', 'y', 'z'} and b = {'c', 'd'}, remove 'c' and print the set.
13 a = {'x', 'y', 'z'}
14 a.remove('c')
15 print("The updated set is: ", a)
16
17 # Given a = {10, 20} and b = {30, 40}, print the union of a and b.
18 a = {10, 20}
19 b = {30, 40}
20 print("The union of a and b is: ", a.union(b))
21
22 # Given a = {1, 2, 3} and b = {4, 5, 6}, print the intersection of a and b.
23 a = {1, 2, 3}
24 b = {4, 5, 6}
25 print("The intersection of a and b is: ", a.intersection(b))
26
27 # Check if 'x' is in the set {'a', 'b', 'c'} and print the result.
28 a = {'a', 'b', 'c'}
29 print("Is 'x' in the set? ", 'x' in a)
```

Output:

```
File Edit Selection View Go Run Python 3.8
PROBLEMS OUTPUT DEBUG CONSOLE PORTS
TERMINAL
PS C:\Users\Sasham Patra\OneDrive\ITR Python\05> python -u "c:\Users\Sasham Patra\OneDrive\ITR Python\05\Tasks 12-06\data_structure\set.py"
The set is: {'a', 'b', 'c'}
The updated set is: {'x', 'y', 'c'}
The updated set is: {'a', 'c'}
The union of a and b is: {10, 20, 30}
The intersection of a and b is: {6}
Is 'x' in the set? True
PS C:\Users\Sasham Patra\OneDrive\ITR Python\05> 
```